

Lab 1: Getting Started with Docker - oadewusi

- Cloning docker github repo

```
oadewusi@oluSec:~/AIOPs/Assign1_docker$ git clone https://github.com/docker/getting-started.git
Cloning into 'getting-started'...
remote: Enumerating objects: 980, done.
remote: Counting objects: 100% (13/13), done.
remote: Compressing objects: 100% (12/12), done.
remote: Total 980 (delta 5), reused 5 (delta 1), pack-reused 967 (from 1)
Receiving objects: 100% (980/980), 5.28 MiB | 1.54 MiB/s, done.
```

- Following docker tutorials by running **docker run -d -p 80:80 docker/getting-started**

```
oadewusi@oluSec:~/AIOPs/Assign1_docker/getting-started$ docker run -d -p 80:80 docker/getting-started
Unable to find image 'docker/getting-started:latest' locally
latest: Pulling from docker/getting-started
c158987b0551: Pull complete
1a35f6679f5b: Pull complete
```

- Creating a Dockerfile to run Our App

```
Dockerfile
Dockerfile > ...
1 FROM node:18-alpine
2 WORKDIR /app
3 COPY . .
4 RUN yarn install --production
5 CMD ["node", "src/index.js"]
6
```

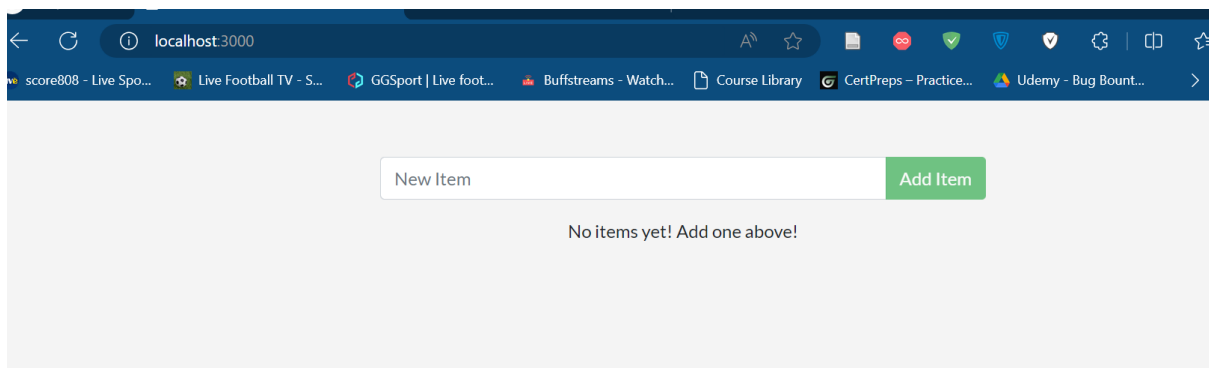
- Building the app from Dockerfile

```
oadewusi@oluSec:~/AIOPs/Assign1_docker/app$ docker build -t getting-started .
[+] Building 21.9s (10/10) FINISHED                                docker:default
=> [internal] load build definition from Dockerfile                0.0s
=> => transferring dockerfile: 137B                                0.0s
=> [internal] load metadata for docker.io/library/node:18-alpine  2.6s
=> [auth] library/node:pull token for registry-1.docker.io        0.0s
=> [internal] load .dockerignore                                   0.0s
=> => transferring context: 2B                                       0.0s
=> [1/4] FROM docker.io/library/node:18-alpine@sha256:02376a266c84acbf45bd19440e08e48b1c8b98037417334046029ab585de0 0.0s
=> [internal] load build context                                   0.1s
=> => transferring context: 4.60MB                                   0.1s
=> CACHED [2/4] WORKDIR /app                                       0.0s
=> [3/4] COPY . .                                                  0.1s
```

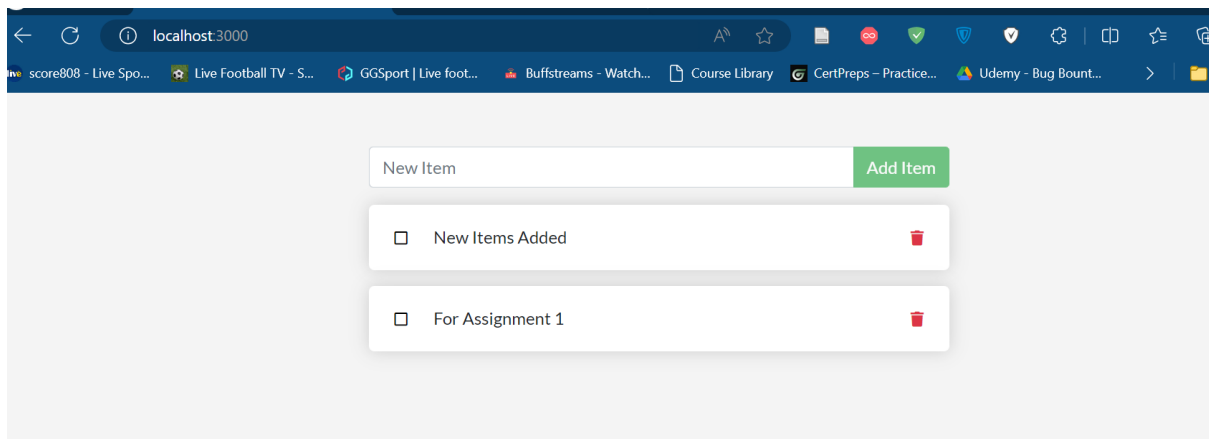
- Starting the App Container

```
oadewusi@oluSec:~/AIOPs/Assign1_docker/app$ docker run -dp 3000:3000 getting-started
f6730e4cab66679d0738c24016fa71e03785cef5fd892c09e94989bbf1a458d4
```

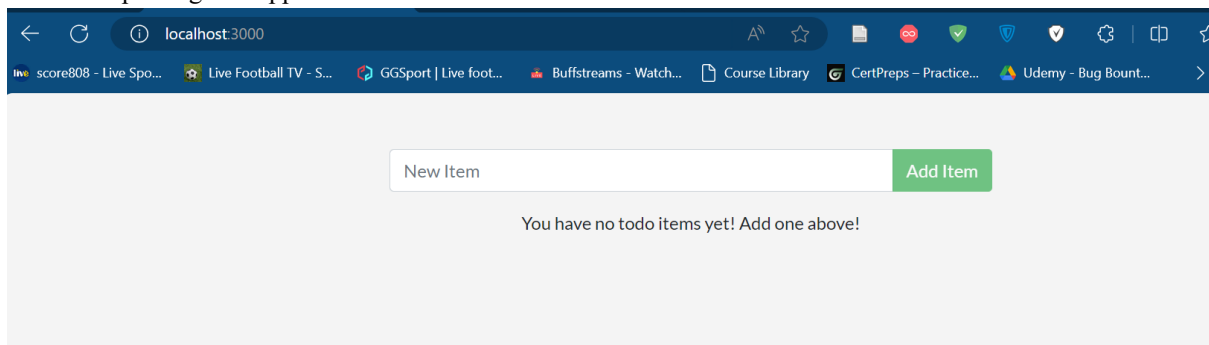
- Verifying the App at <http://localhost:3000>



- Adding Items to the App



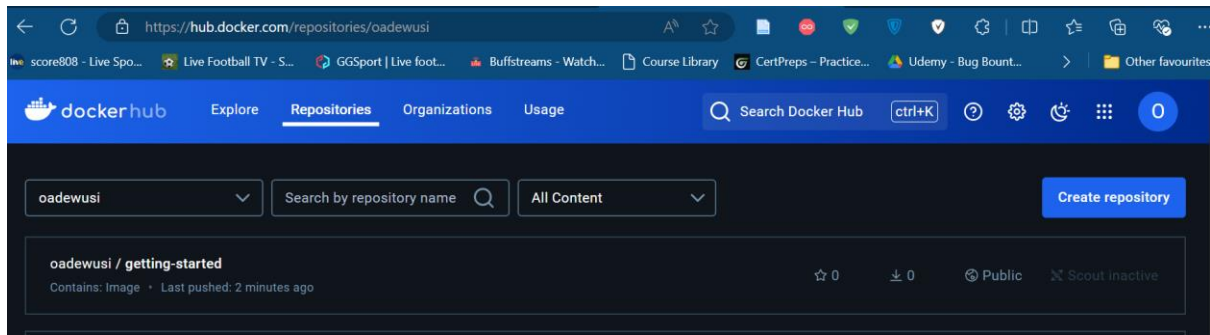
- Updating our App



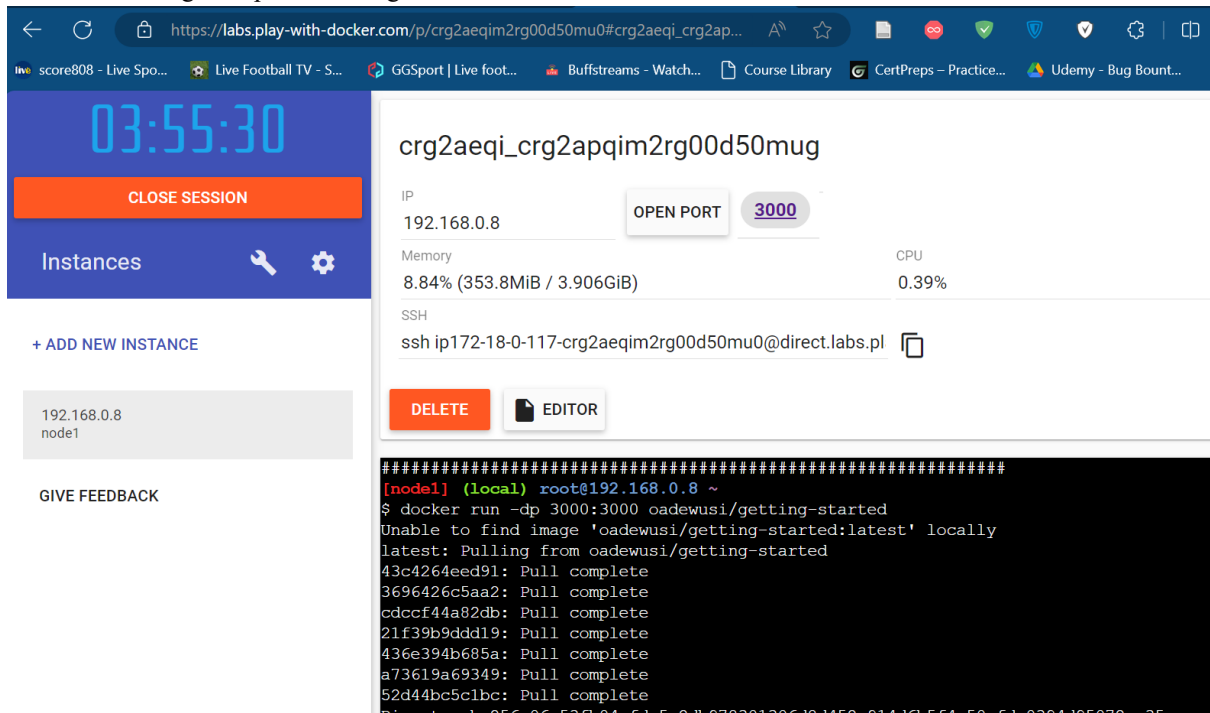
- Restarting the container after the update

```
oadevusi@oluSec:~/AIOPs/Assign1_docker/app$ docker run -dp 3000:3000 getting-started
992dd62f2cc87a30d34c9145a2c3f6403f5f5d4892080578437fddf1b6b05c9c
oadevusi@oluSec:~/AIOPs/Assign1_docker/app$ docker tag getting-started oadevusi/getting-started
oadevusi@oluSec:~/AIOPs/Assign1_docker/app$ docker push oadevusi/getting-started
Using default tag: latest
The push refers to repository [docker.io/oadevusi/getting-started]
0632178d65dd: Pushed
b32c22f2f169: Pushed
f543d1adc3d9: Pushed
e2be10e97665: Mounted from library/node
06fd85419b65: Mounted from library/node
f58c462fa079: Mounted from library/node
63ca1fbb43ae: Mounted from library/node
latest: digest: sha256:06e53fb04cfda5e2db978301306d8d452a914d6b5f4c50cfde0394d95072ae35 size: 1787
```

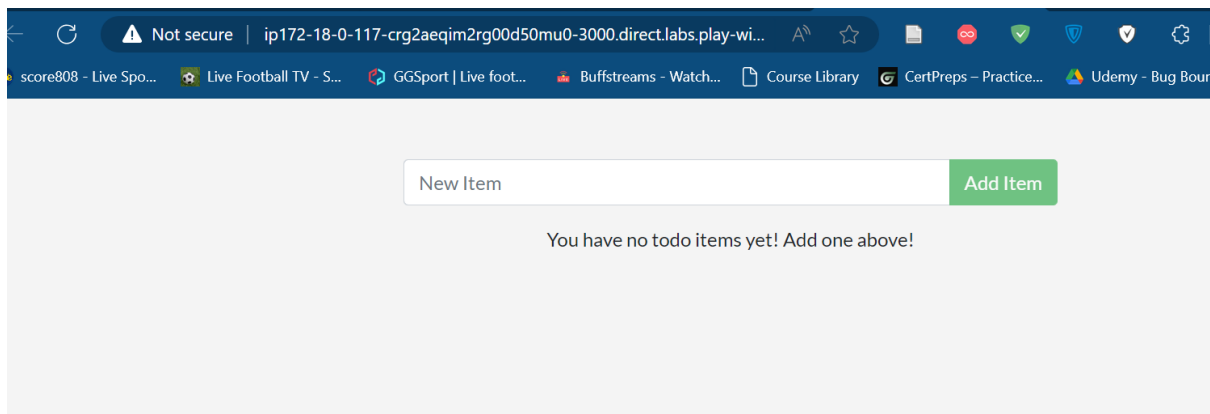
- Sharing our App



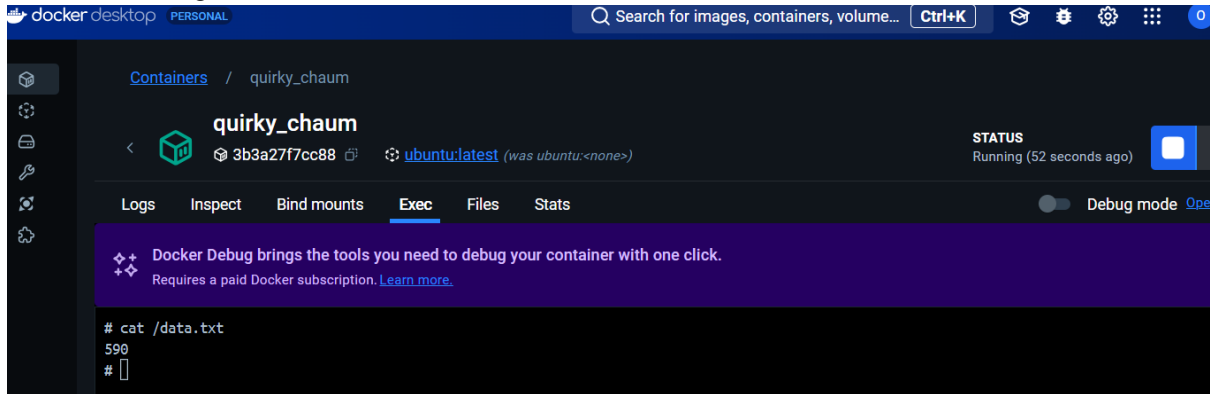
- Running the uploaded Image on a New Instance



- Verification of the instance



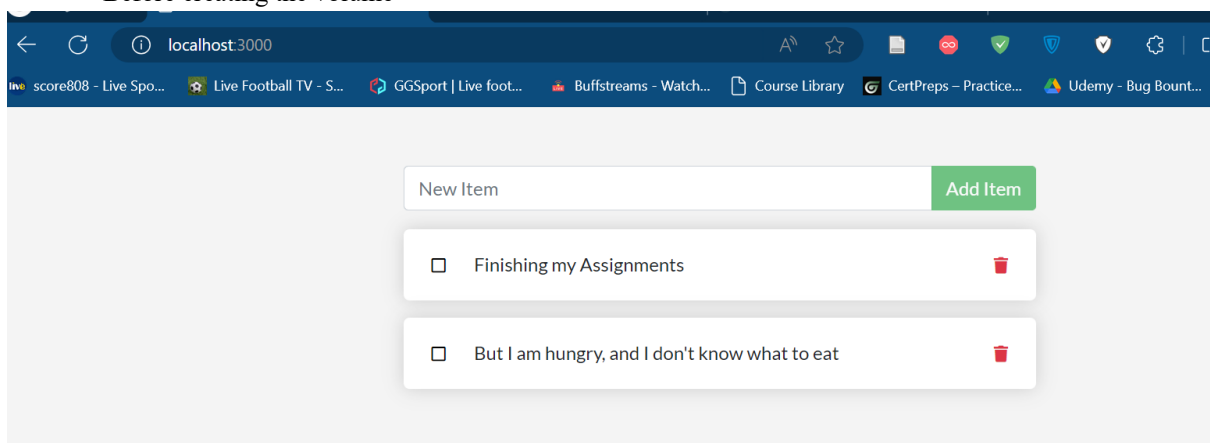
- Persisting our DB



```
oadeuwusi@oluSec:~/AIOPs/Assign1_docker/app$ docker run -it ubuntu
root@f4555a5475c3:/# ls
bin boot dev etc home lib lib64 media mnt opt proc root run sbin srv sys tmp usr var
root@f4555a5475c3:/#
```

```
exit
oadeuwusi@oluSec:~/AIOPs/Assign1_docker/app$ docker volume create todo-db
todo-db
oadeuwusi@oluSec:~/AIOPs/Assign1_docker/app$ docker run -dp 3000:3000 -v todo-db:/etc/todos getting-started
5f8119093d183cc647e1b58e93b88d3c6ce821140b58adb7c00d7ca9dc051fae
```

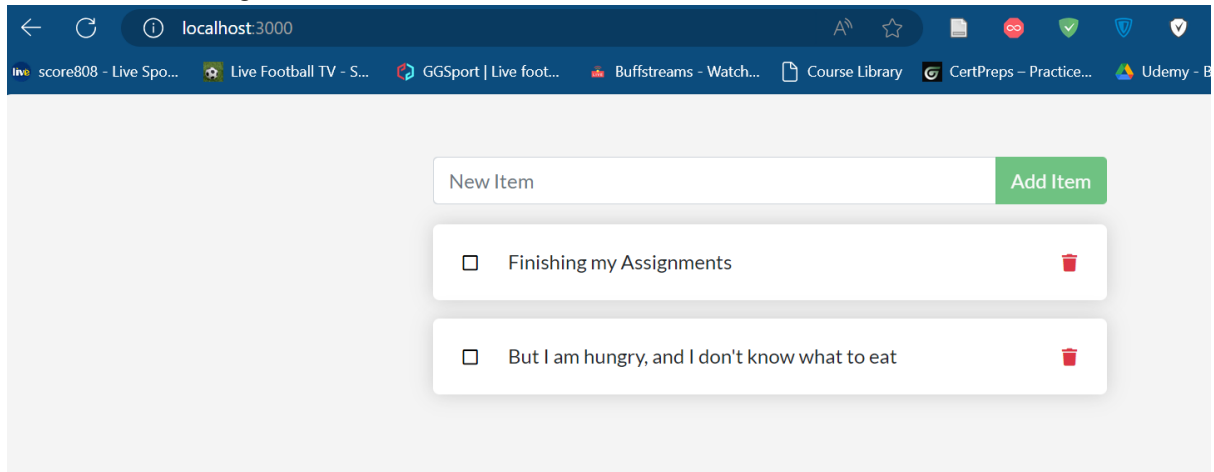
- Creating persistence with a volume
Before creating the volume



- Persisting the Todo Data

```
oadeuwusi@oluSec:~/AIOPs/Assign1_docker/app$ docker ps
CONTAINER ID   IMAGE      COMMAND                  CREATED        STATUS        PORTS
MES           5f8119093d18 getting-started "docker-entrypoint.s..." 2 minutes ago Up 2 minutes  0.0.0.0:3000->3000/tcp
cused_tharp    34ddb9b408 docker/getting-started "/docker-entrypoint..." 3 hours ago   Up 3 hours    0.0.0.0:80->80/tcp
ngry_dewdney
oadeuwusi@oluSec:~/AIOPs/Assign1_docker/app$ docker rm -f 5f811
5f811
oadeuwusi@oluSec:~/AIOPs/Assign1_docker/app$ docker run -dp 3000:3000 -v todo-db:/etc/todos getting-started
5ec154f00cd52f14303f7abb0c31c6c49b43f4f76e06808d8e91c612cb3f3a28
```

- After creating the volume



- Inspecting the volume

```
oadevusi@oluSec:~/AIOPs/Assign1_docker/app$ docker volume inspect todo-db
[
  {
    "CreatedAt": "2024-09-10T11:05:19Z",
    "Driver": "local",
    "Labels": null,
    "Mountpoint": "/var/lib/docker/volumes/todo-db/_data",
    "Name": "todo-db",
    "Options": null,
    "Scope": "local"
  }
]
```

- Created docker compose file with a common network for the app and the DB – app-network

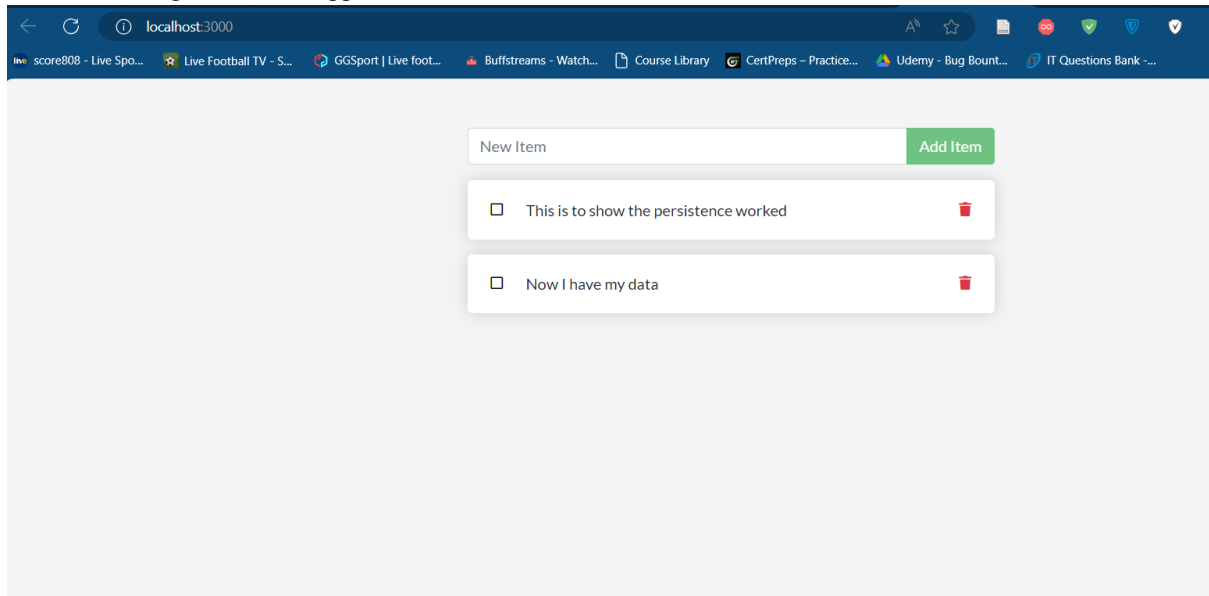
```
docker-compose.yml X Dockerfile

docker-compose.yml
1  services:
2    app:
3      image: oadewusi/getting-started
4      ports:
5        - "3000:3000"
6      working_dir: /app
7      environment:
8        MYSQL_HOST: mysql
9        MYSQL_USER: root
10       MYSQL_PASSWORD: secret
11       MYSQL_DB: todos
12     networks:
13       - app-network
14
15     mysql:
16       image: mysql:8.0
17       volumes:
18         - todo-mysql-data:/var/lib/mysql
19       environment:
20         MYSQL_ROOT_PASSWORD: secret
21         MYSQL_DATABASE: todos
22       hostname: mysql
23       networks:
24         - app-network
```

- Started the containers.

```
oadewusi@oluSec:~/AIOPs/Assign1_docker/app$ docker compose up -d
[+] Running 12/1
✓ mysql Pulled
[+] Running 4/4
✓ Network app_app-network Created
✓ Volume "app_todo-mysql-data" Created
✓ Container app-mysql-1 Started
✓ Container app-app-1 Started
```

- Adding Item to the App



- Inspecting the app-network

```
oadevusio@oluSec:~/AIOPs/Assign1_docker/app$ docker network inspect 352f55739eed
[
  {
    "Name": "app_app-network",
    "Id": "352f55739eed4441c228ff8dc7ae31003c043d67efcfac8a657984519299513f",
    "Created": "2024-09-10T12:07:35.839052718Z",
    "Scope": "local",
```

```
  },
  "Internal": false,
  "Attachable": false,
  "Ingress": false,
  "ConfigFrom": {
    "Network": ""
  },
  "ConfigOnly": false,
  "Containers": {
    "7a23acd6270db9a246982c09e0b88b3921383080215fb86bf39c176754900640": {
      "Name": "app-app-1",
      "EndpointID": "446da26437cab2969439826faf24a9014d76e20111f8122cb44c0416b82f3556",
      "MacAddress": "02:42:ac:12:00:03",
      "IPv4Address": "172.18.0.3/16",
      "IPv6Address": ""
    },
    "9cd4431ef4b8016849ea35052c7f291bdfbe57b5cf8ad702ffc62c4666d17d90": {
      "Name": "app-mysql-1",
      "EndpointID": "991887c1fefcc1b914c8f78d87989001f1dc9f9ec92925c1992044d54249c2a5",
      "MacAddress": "02:42:ac:12:00:02",
      "IPv4Address": "172.18.0.2/16",
      "IPv6Address": ""
    }
  },
  "Options": {},
  "Labels": {
    "com.docker.compose.network": "app-network",
    "com.docker.compose.project": "app",
    "com.docker.compose.version": "2.29.2"
  }
}
```

- Checked the logs to see if the DB mysql is connected.

```
oadeuwusi@oluSec:~/AIOPs/Assign1_docker/app$ docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS                    NAMES
9cd4431ef4b8   mysql:8.0     "docker-entrypoint.s..." 2 hours ago   Up 2 hours   3306/tcp, 33060/tcp      app-mysql-1
7a23acd6270d   oadeuwusi/getting-started "docker-entrypoint.s..." 2 hours ago   Up 2 hours   0.0.0.0:3000->3000/tcp    app-app-1
34ddbcc9b408   docker/getting-started "/docker-entrypoint..." 7 hours ago   Up 7 hours   0.0.0.0:80->80/tcp        hungry_dewdney

oadeuwusi@oluSec:~/AIOPs/Assign1_docker/app$ docker logs 7a23acd6270d
Waiting for mysql:3306.....
Timeout
Error: connect ECONNREFUSED 172.18.0.2:3306
    at TCPConnectWrap.afterConnect [as oncomplete] (node:net:1555:16) {
  errno: -111,
  code: 'ECONNREFUSED',
  syscall: 'connect',
  address: '172.18.0.2',
  port: 3306,
  fatal: true
}
Waiting for mysql:3306.
Connected!
Connected to mysql db at host mysql
Listening on port 3000
```

- Replaced the CMD command on the Dockerfile as instructed.

```
Dockerfile X
app > Dockerfile > FROM
1 FROM node:18-alpine
2 WORKDIR /app
3 COPY . .
4 RUN yarn install --production
5 CMD ["tail", "-f", "/dev/null"]
6
```


- Rebuilding the image and the container using the new compose file

```

docker-compose.yml X package.json Dockerfile
docker-compose.yml
1  services:
2    app:
3      image: oadewusi/getting-started
4      ports:
5        - "3000:3000"
6      working_dir: /app
7      # environment:
8      #   MYSQL_HOST: mysql
9      #   MYSQL_USER: root
10     #   MYSQL_PASSWORD: secret
11     #   MYSQL_DB: todos
12     networks:
13       - app-network
14
15     mysql:
16       image: mysql:8.0
17       volumes:
18         - todo-mysql-data:/var/lib/mysql
19       environment:
20         MYSQL_ROOT_PASSWORD: secret
21         MYSQL_DATABASE: todos
22       hostname: mysql

```

```

oadewusi@oluSec:~/AIOPs/Assign1_docker/app$ docker exec -it 7a25ac0270d sh
/app # oadewusi@oluSec:~/AIOPs/Assign1_docker$ docker build -t getting-started .
[+] Building 24.9s (10/10) FINISHED
      docker:default
=> [internal] load build definition from Dockerfile
      0.0s
=> => transferring dockerfile: 141B
      0.0s

```

```

oadewusi@oluSec:~/AIOPs/Assign1_docker/app$ docker compose up -d
[+] Running 7/9
  :: mysql Pulling
  :: app [██████████] Pulling
    ✓ 43c4264eed91 Already exists
    ✓ 3696426c5aa2 Already exists
    ✓ cdccf44a82db Already exists
    ✓ 21f39b9ddd19 Already exists

```

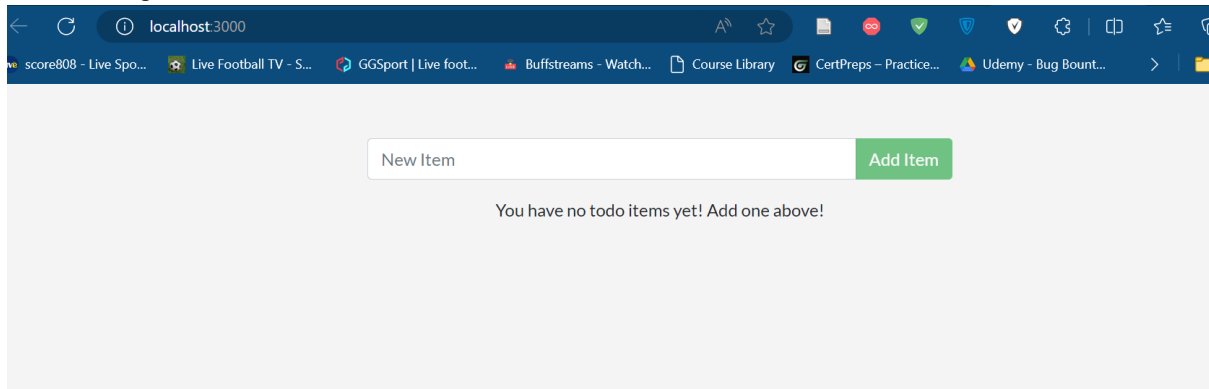
- Remote in to the running container

```

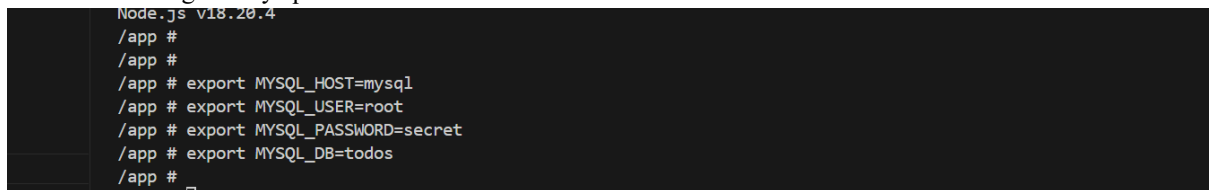
oadewusi@oluSec:~/AIOPs/Assign1_docker/app$ docker exec -it fae0a6ce1d8c sh
/app # node src/index.js
Using sqlite database at /etc/todos/todo.db
node:events:495

```

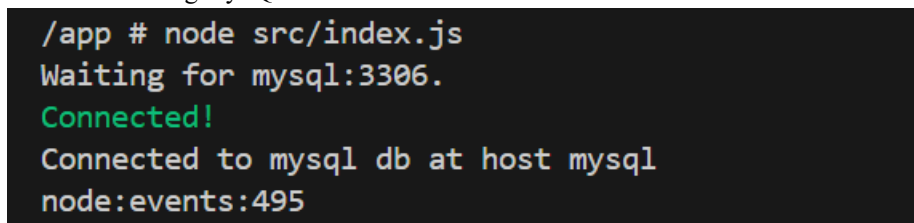
- No persistence



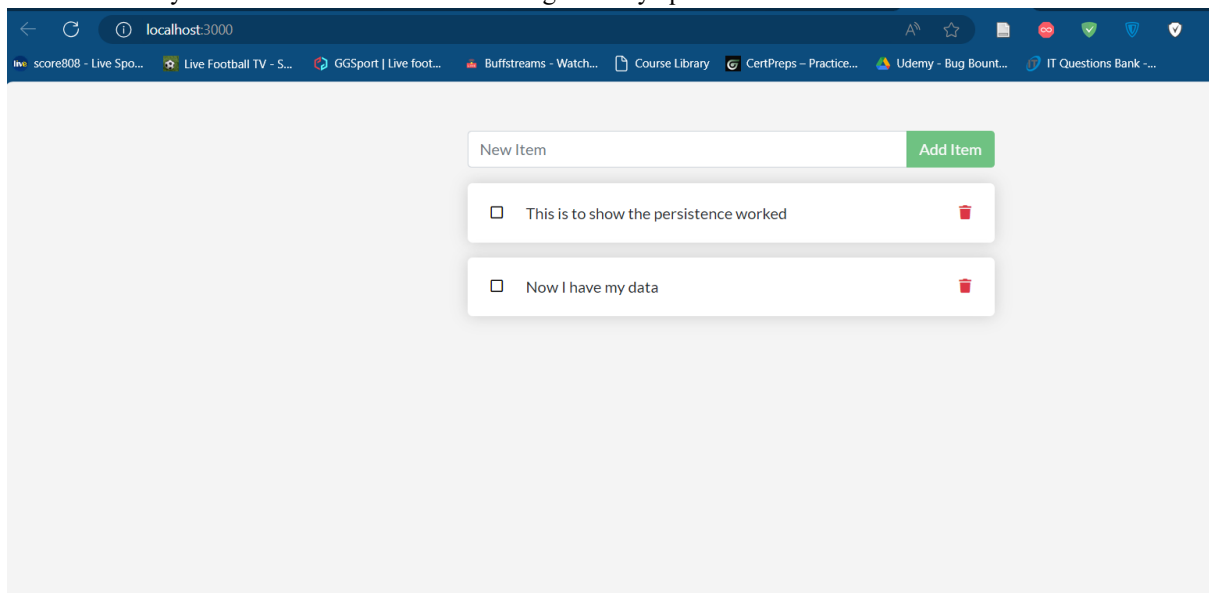
- Entering the mysql env variables in the shell



- Showing mySQL connected



- Got my added Items back after connecting with mysql

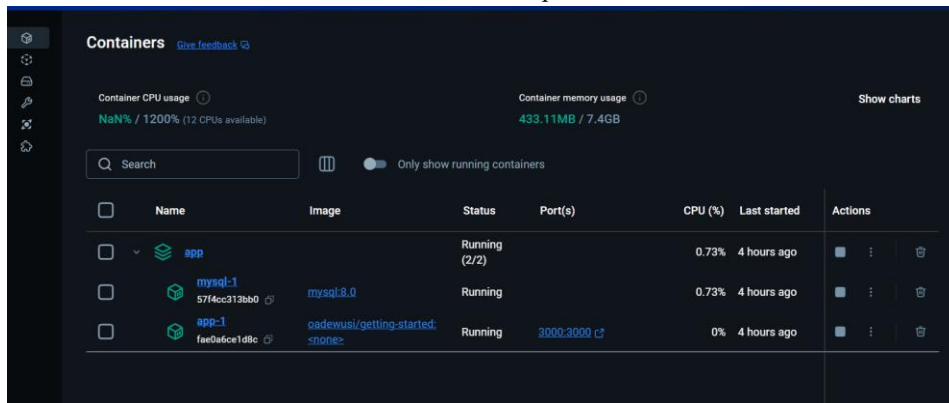


- Confirming the volume created

```
oadevusi@oluSec:~/AIOPs/Assign1_docker/app$ docker volume ls
DRIVER      VOLUME NAME
local       app_todo-mysql-data
local       todo-db
```

```
oadevusi@oluSec:~/AIOPs/Assign1_docker/app$ docker volume inspect app_todo-mysql-data
[
  {
    "CreatedAt": "2024-09-10T12:07:36Z",
    "Driver": "local",
    "Labels": {
      "com.docker.compose.project": "app",
      "com.docker.compose.version": "2.29.2",
      "com.docker.compose.volume": "todo-mysql-data"
    },
    "Mountpoint": "/var/lib/docker/volumes/app_todo-mysql-data/_data",
    "Name": "app_todo-mysql-data",
    "Options": null,
    "Scope": "local"
  }
]
```

- Containers view from the Docker Desktop



- Volume view from the Docker Desktop

